

Second part: Open book

Total score: 10 pts.

You are warmly invited to be tidy and concise. Messy papers will be penalized.

6. A soccer team has to upgrade the stadium before the beginning of the championship. The upgrade project is decomposed into n activities $A = \{1, \dots, n\}$. Each activity j has a nominal processing time of p_j days. The set of oriented pairs $P \subset A \times A$ gives the precedence between pairs of activities (i.e., if $(i, j) \in P$ then i must finish before j begins). Activity 1 precedes all the other ones and activity n follows all the other ones, both activities have a symbolic duration of 0 days.

Formulate the problem of deciding the starting time of all activities, compatible with the precedence constraints, that minimizes the total duration of the project (difference between the starting times of n and 1) as a linear optimization problem.

Variante: Since the duration of the project is too long and the works will finish after the beginning of the championship, the team manager decides to give money incentives to the working company in order to speed up the works. For every activity j , p'_j ($< p_j$) is the minimum allowable working time, and c_j is the cost for each saved day. Let D be the desired maximum duration of the project. The problem now is to decide the duration of each activity (or the number of saved days per activity) so that the overall duration is equal to D and the cost for the incentives is minimized. Describe how to change the model.

7. A Corsican honey producer owns some transportable beehives. The producer is used to place the beehives in the fields where flowers are blooming. There are n fields. For every field i it is known the day when the blooming is starting (a_i), the day when the blooming is ending (b_i) and the revenue from the honey produced by the bees in that field (r_i). The producer has to decide which fields to visit in order to maximize the total revenue. Only one field can be visited at a time (it is not possible to place part of the beehives in one field and part in another one), and in order to get the revenue for a given field the bees must remain there for the entirety of the blooming (thus for field i the bees must remain there from a_i to b_i).

Formulate the problem as the search of a maximum or minimum path on a suitable graph. Describe the construction of the graph, the meaning of the nodes and of the arcs, the cost/length of the arcs. What is the main property of the graph? What is the most efficient algorithm to be used to solve the problem? Justify your answers.